

Machine Learning

Assignment 3

Report

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Objective

The goal of this assignment is to implement and evaluate deep learning algorithms on the MNIST dataset using PyTorch. The experiments include training Multi-Layer Perceptrons (MLPs) and Convolutional Neural Networks (CNNs), comparing their performance with a classical machine learning model used in Assignment 1.

Dataset and Frameworks

- **Dataset:** MNIST (handwritten digits, 28×28 grayscale images)
- **Framework:** PyTorch
- **Metrics:**
 - Accuracy on test set
 - Training time in seconds

Task A: Multi-Layer Perceptron (MLP)

Task Description		Accuracy	Time (s)
A.1	3-layer MLP (ReLU + Adam)	97.9%	45
A.2	3-layer MLP (Sigmoid + Adam)	95.8%	48
A.3	3-layer MLP (ReLU + SGD)	96.3%	51
A.4	5-layer MLP (ReLU + Adam) – Best MLP	98.2%	65

Findings:

- ReLU activation performed significantly better than Sigmoid.
- The Adam optimizer yielded faster convergence and higher accuracy than SGD.
- Adding depth (5 layers) improved model performance in A.4.

Task B: Convolutional Neural Networks (CNNs)

Task	Model Used	Accuracy	Time (s)
B.1	4-layer custom CNN	98.6%	55
B.2	LeNet-5	98.3%	40

Choice & Justification for B.2:

For Task B.2, I selected **LeNet-5**, a foundational CNN model developed specifically for digit recognition. Its architecture is well-suited for the MNIST dataset and offers a great balance of performance and efficiency. It trained quickly (under 2 minutes) and still achieved over 98% accuracy.

Task C: Performance Comparison and Analysis

Model	Accuracy	Time (s)	Notes
XGBoost (Assgn 1)	97.2%	30	Best classical ML model
MLP (A.4)	98.2%	65	Best MLP
LeNet-5 (B.2)	98.3%	40	Best standard CNN model
4-layer CNN (B.1)	98.6%	55	Best overall deep model

Conclusion

The best-performing model overall was the **4-layer CNN from Task B.1**, achieving the highest accuracy at 98.6% while staying well within runtime constraints. **LeNet-5**, chosen for Task B.2, offered an excellent compromise between speed and accuracy. Compared to traditional models like XGBoost (used in Assignment 1), CNNs showed clear advantages in handling image-based data.

Best Overall Model: 4-layer CNN (B.1)

Acknowledgment of Sources

Throughout this assignment, I referred to a variety of online resources, including Google, ChatGPT, and open-source documentation. These tools helped with debugging, understanding deep learning architecture decisions, and speeding up model development and testing.